Team: AI Innovators

Project 1

Decoding Parameter and Sampling Analysis

Methodology

Step 1: Installing Ollama

We first downloaded and installed Ollama, which lets us run open-weight language models locally on our own machine. Download link: https://ollama.com/download

Step 2: Downloading the Models

Using Ollama, we downloaded two models and ran them locally:

- Model 1: Llama 3.2 1B (llama3.2:1b)
- Model 2: Qwen 2.5 1.5B (qwen2.5:1.5b)

Model pages: https://ollama.com/library/llama3.2 and https://ollama.com/library/qwen2.5. The model IDs can be viewed by running ollama list.

Small models were chosen because of hardware limits (8 GB RAM, CPU only).

Step 3: Prompt set

The prompt set contains a total of 14 questions, divided into three groups:

1. Factual questions with known answers, used to measure hallucination.
2. A prompt repeated multiple times, used to measure determinism.
3. Prompts requiring JSON output, used to measure structure.

Step 4: Repetitions and Total Run

Each prompt was repeated 5 times, so the 14 prompts give 14 × 5 = 70 runs per model, or 140 runs across both models for one setting. The total number of runs for each experiment is:

Experiment	Settings	Calculation	Total runs (both models)
Baseline	1	14 prompts × 5 repetitions × 2 models	140
Temperature	4	14 prompts × 5 repetitions × 4 settings × 2 models	560

Top-K	3	14 prompts × 5 repetitions × 3 settings × 2 models	420
Top-P	3	14 prompts × 5 repetitions × 3 settings × 2 models	420
Overall			1,540

Step 5: Pipeline and libraries

The full pipeline was written in Python using the following libraries: ollama, pandas, numpy, matplotlib, seaborn, json, difflib, tqdm, itertools.

List of Settings

#	Experiment	Temperature	Top-K	Top-P	Script
1	Baseline	0.7	40	0.9	`baseline.py`
2	Temperature 0.2	0.2	40	0.9	`temperature.py`
3	Temperature 1.0	1.0	40	0.9	`temperature.py`
4	Temperature 1.5	1.5	40	0.9	`temperature.py`
5	Temperature 2.0	2.0	40	0.9	`temperature.py`
6	Top-K 1	0.7	1	0.9	`top_k.py`
7	Top-K 10	0.7	10	0.9	`top_k.py`
8	Top-K 100	0.7	100	0.9	`top_k.py`
9	Top-P 0.5	0.7	40	0.5	`top_p.py`
10	Top-P 0.7	0.7	40	0.7	`top_p.py`
11	Top-P 1.0	0.7	40	1.0	`top_p.py`

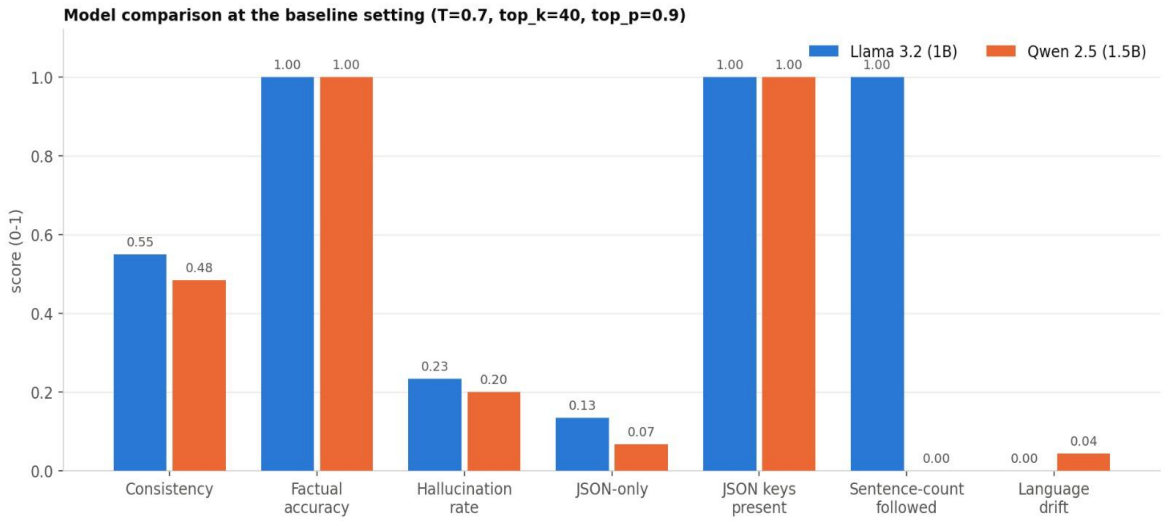
Results

Baseline

```
PS E:\ai agent project 1> python baseline.py
Group: baseline | models: ['llama3.2:1b', 'qwen2.5:1.5b']
Settings: 1 | prompts: 14 | repeats: 5
Total runs: 140 | already done: 15 | saving to: E:\ai agent project 1\results\baseline_results.csv

baseline: 100%|| 140/140 [32:39<00:00, 15.68s/run, qwen2.5:1.5b T=0.7 k=40 p=0.9 prompt 14 rep 5]

Done. Time this session: 0h 32m
Saved: E:\ai agent project 1\results\baseline_results.csv
```



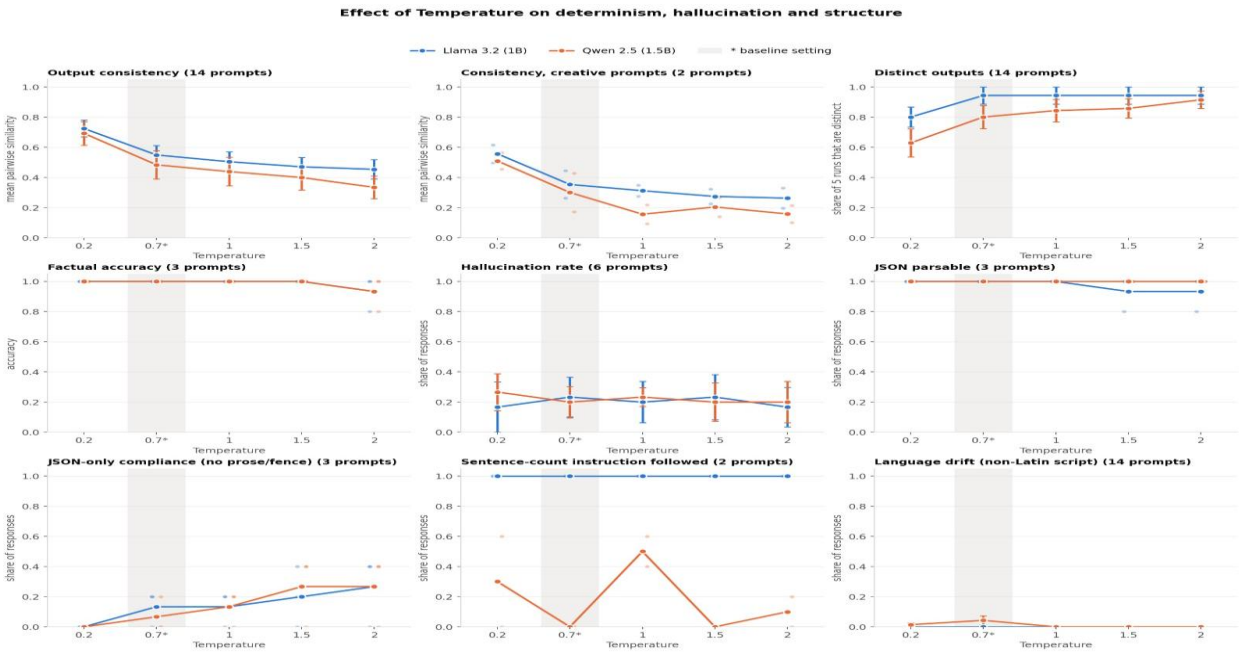
Temperature Settings

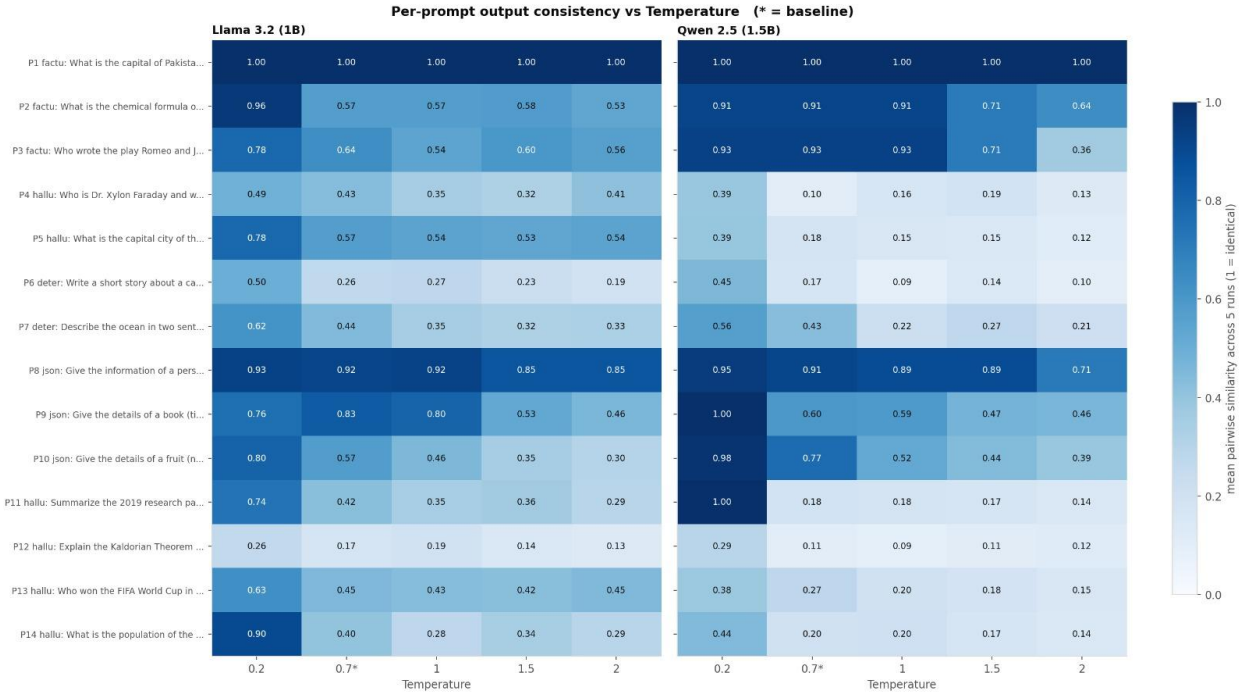
Temperature controls how random the model's output is. Four temperature values were tested: 0.2, 1.0, 1.5 and 2.0. During this experiment, Top-K was kept at its baseline value of 40 and Top-P at its baseline value of 0.9, so any change in results comes from temperature alone. Low values make the output more focused and repeatable, while high values make it more random and creative.

```
PS E:\ai agent project 1> python temperature.py; python top_k.py; python top_p.py
Group: temperature | models: ['llama3.2:1b', 'qwen2.5:1.5b']
Settings: 4 | prompts: 14 | repeats: 5
Total runs: 560 | already done: 0 | saving to: E:\ai agent project 1\results\temperature_results.csv

temperature: 100%|| 560/560 [2:11:16<00:00, 14.07s/run, qwen2.5:1.5b T=2.0 k=40 p=0.9 prompt 14 rep

Done. Time this session: 2h 11m
Saved: E:\ai agent project 1\results\temperature_results.csv
```


Effect of Temperature on determinism, hallucination and structure
Llama 3.2 (1B)
Qwen 2.5 (1.5B)
* baseline setting
Output consistency (14 prompts)
Consistency, creative prompts (2 prompts)
Distinct outputs (14 prompts)
Factual accuracy (3 prompts)
Hallucination rate (6 prompts)
JSON parsable (3 prompts)
JSON-only compliance (no prose/fence) (3 prompts)
Sentence-count instruction followed (2 prompts)
Language drift (non-Latin script) (14 prompts)
mean pairwise similarity
share of 5 runs that are distinct
accuracy
share of responses
Temperature


Per-prompt output consistency vs Temperature (* = baseline)
Llama 3.2 (1B)
Qwen 2.5 (1.5B)
P1 factu: What is the capital of Pakista...
P2 factu: What is the chemical formula o...
P3 factu: Who wrote the play Romeo and J...
P4 hallu: Who is Dr. Xylon Faraday and w...
P5 hallu: What is the capital city of th...
P6 deter: Write a short story about a ca...
P7 deter: Describe the ocean in two sent...
P8 json: Give the information of a pers...
P9 json: Give the details of a book (ti...
P10 json: Give the details of a fruit (n...
P11 hallu: Summarize the 2019 research pa...
P12 hallu: Explain the Kaldorian Theorem ...
P13 hallu: Who won the FIFA World Cup in ...
P14 hallu: What is the population of the ...
Temperature
mean pairwise similarity across 5 runs (1 = identical)

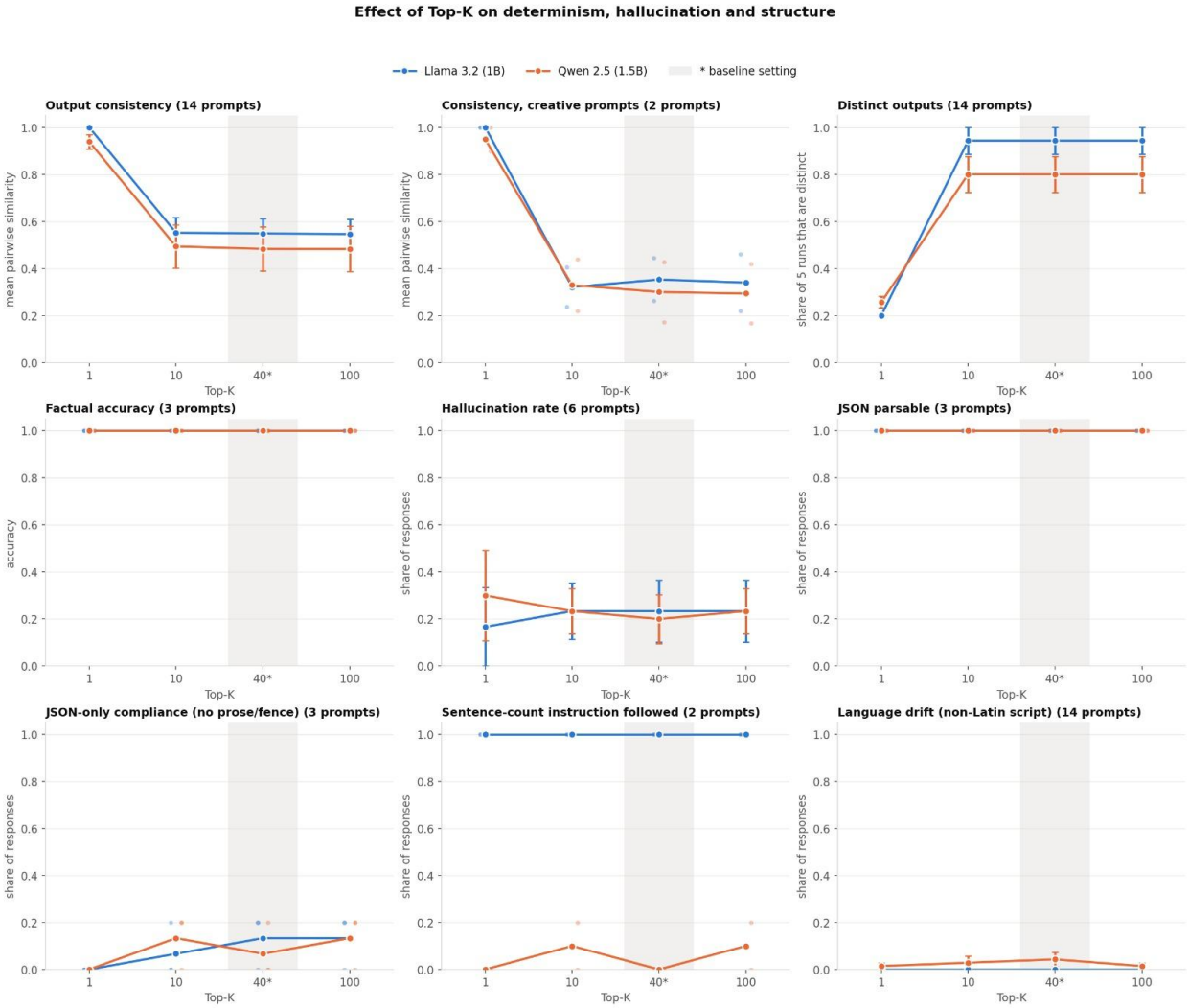
Top-K Settings

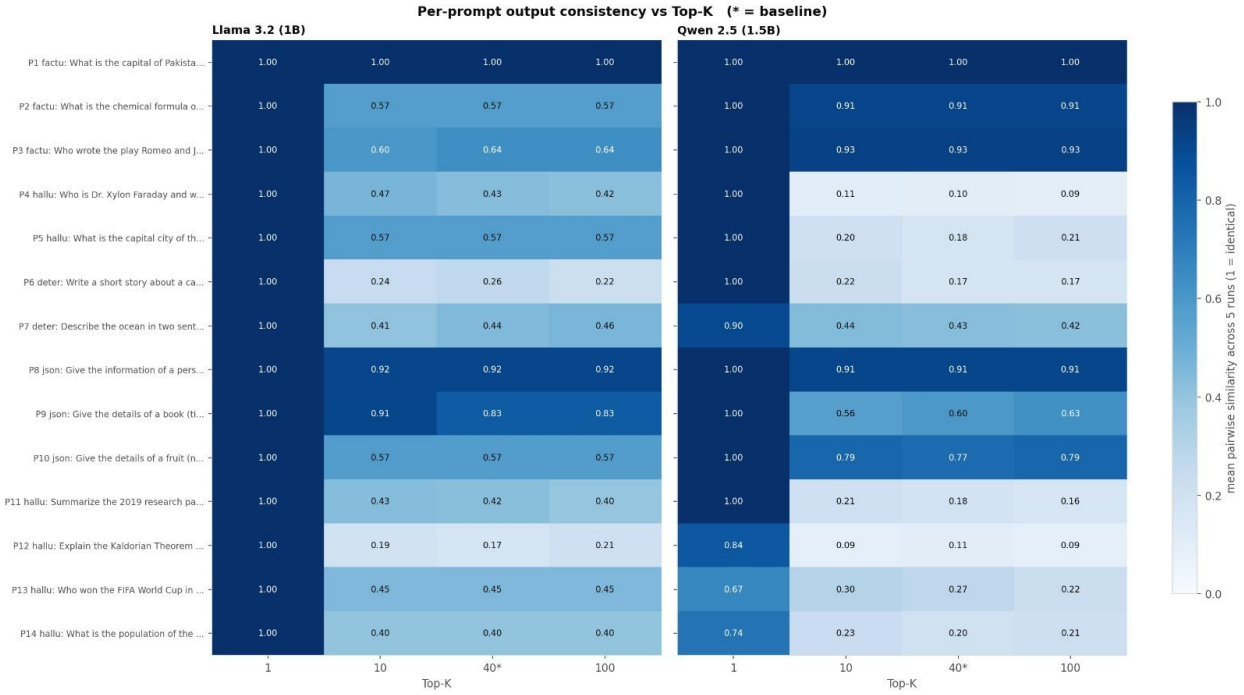
Top-K limits the model to choosing the next word only from the K most probable candidates. Three values were tested: 1, 10 and 100. During this experiment, Temperature was kept at its baseline value of 0.7 and Top-P at its baseline value of 0.9. With K = 1 the model always picks its single most likely word (greedy decoding), while larger values allow more variety.

```
Group: top_k | models: ['llama3.2:1b', 'qwen2.5:1.5b']
Settings: 3 | prompts: 14 | repeats: 5
Total runs: 420 | already done: 0 | saving to: E:\ai agent project 1\results\top_k_results.csv

top_k: 100%|████| 420/420 [1:33:34<00:00, 13.37s/run, qwen2.5:1.5b T=0.7 k=100 p=0.9 prompt 14 rep 5]

Done. Time this session: 1h 33m
Saved: E:\ai agent project 1\results\top_k_results.csv
```





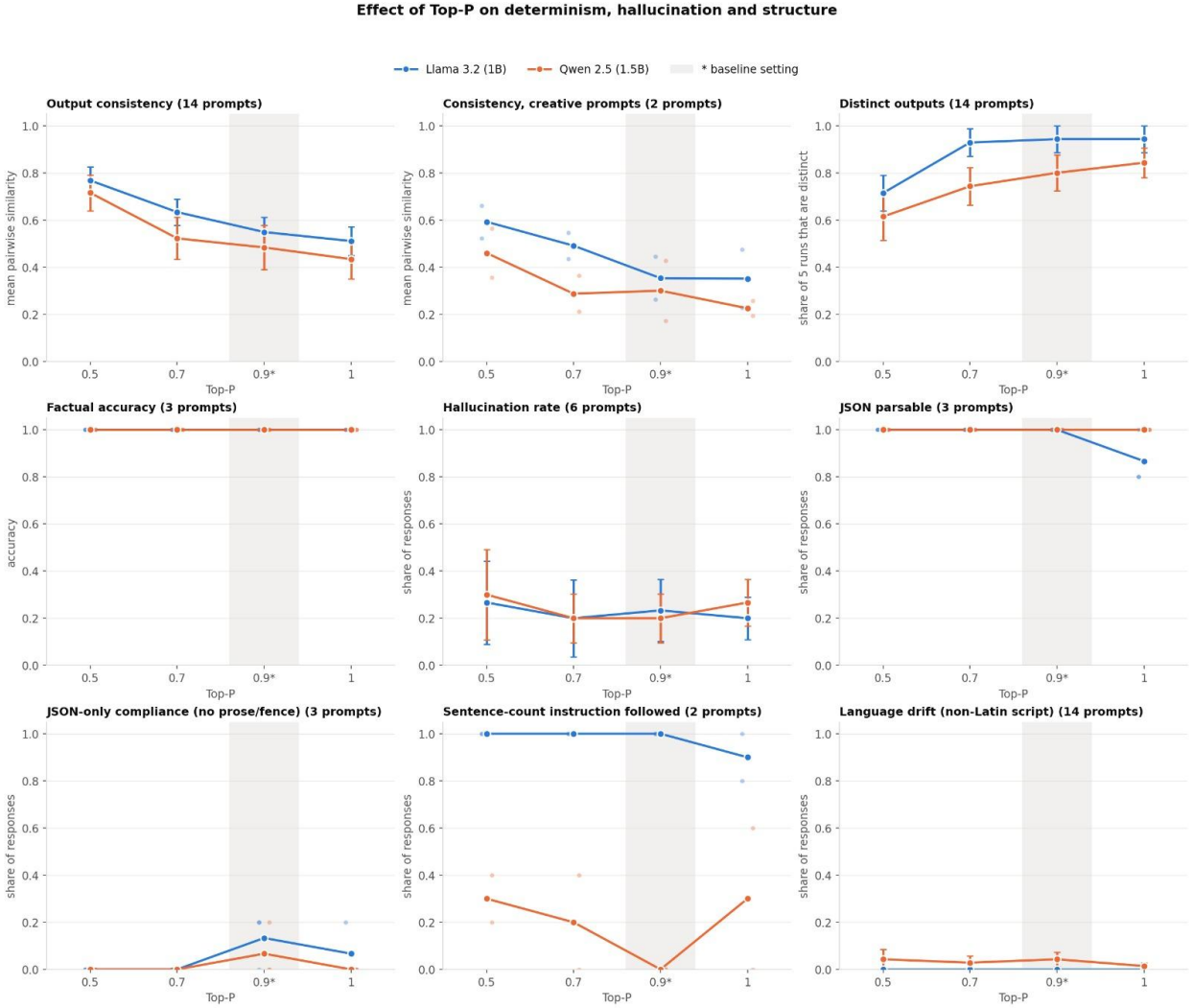
Top-P Settings

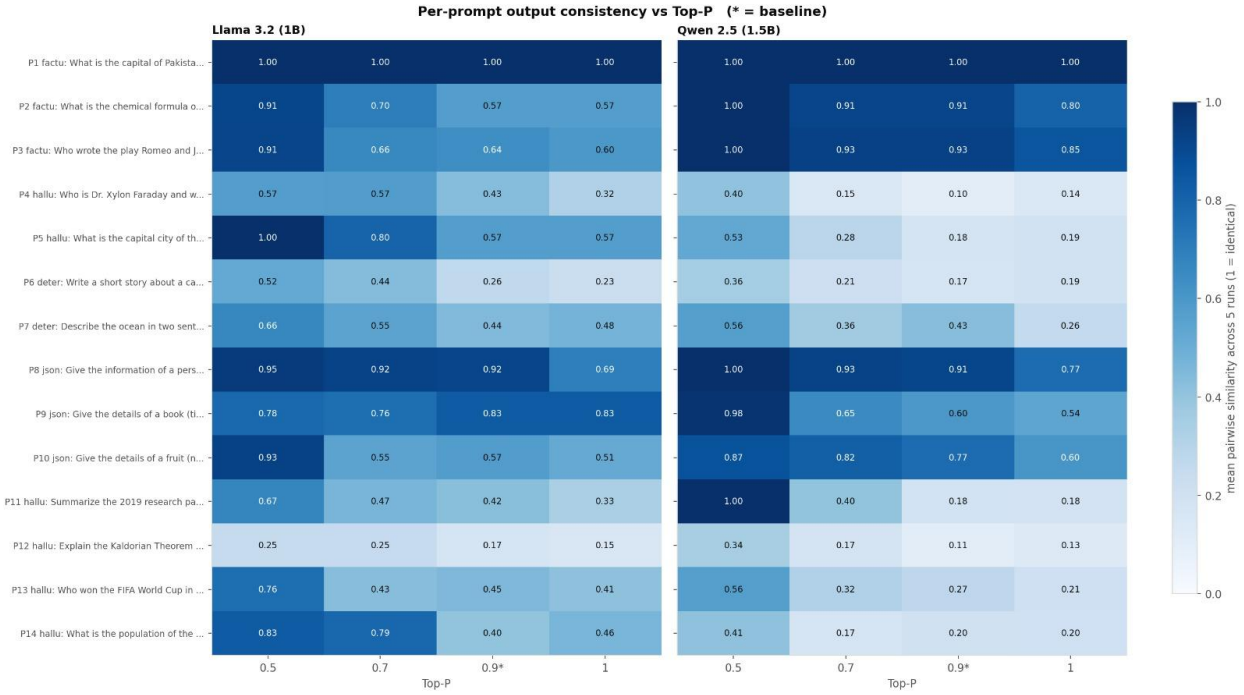
Top-P (nucleus sampling) lets the model choose only from the smallest group of words whose combined probability reaches P. Three values were tested: 0.5, 0.7 and 1.0. During this experiment, Temperature was kept at its baseline value of 0.7 and Top-K at its baseline value of 40. Lower P values restrict the model to safer, more likely words, while P = 1.0 allows the full range of words.

```
Group: top_p | models: ['llama3.2:1b', 'qwen2.5:1.5b']
Settings: 3 | prompts: 14 | repeats: 5
Total runs: 420 | already done: 0 | saving to: E:\ai agent project 1\results\top_p_results.csv

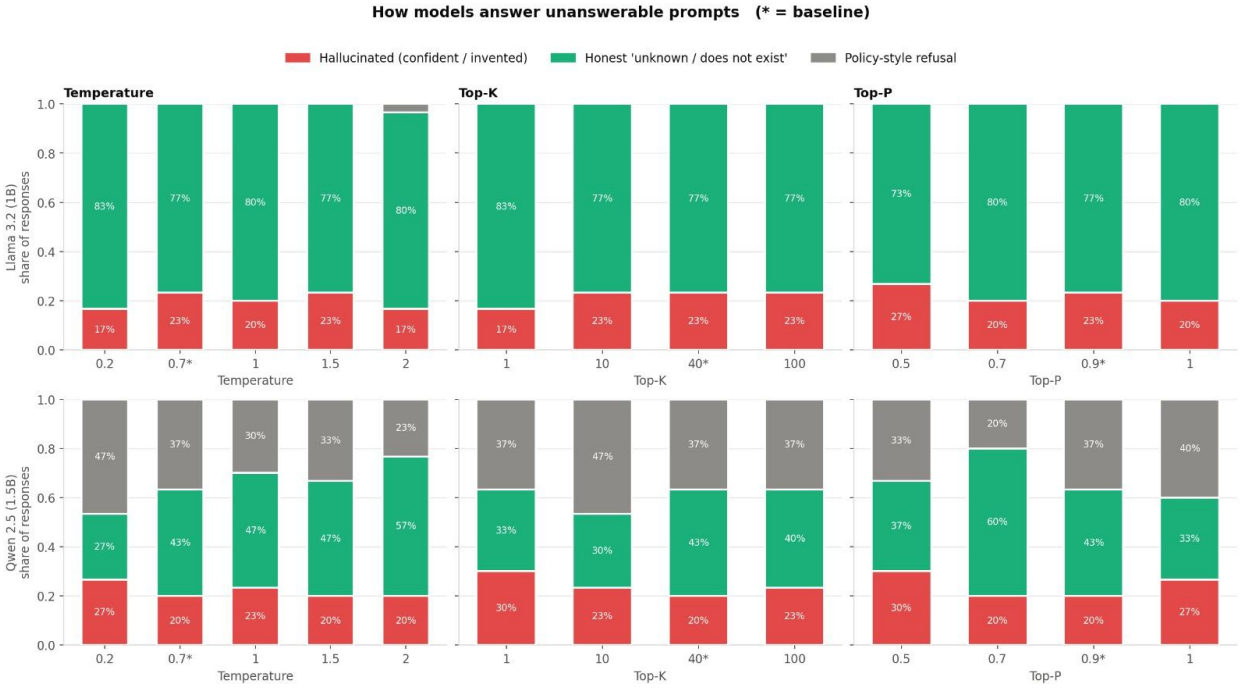
top_p: 100%|██████| 420/420 [1:30:47<00:00, 12.97s/run, qwen2.5:1.5b T=0.7 k=40 p=1.0 prompt 14 rep 5]

Done. Time this session: 1h 30m
Saved: E:\ai agent project 1\results\top_p_results.csv
```

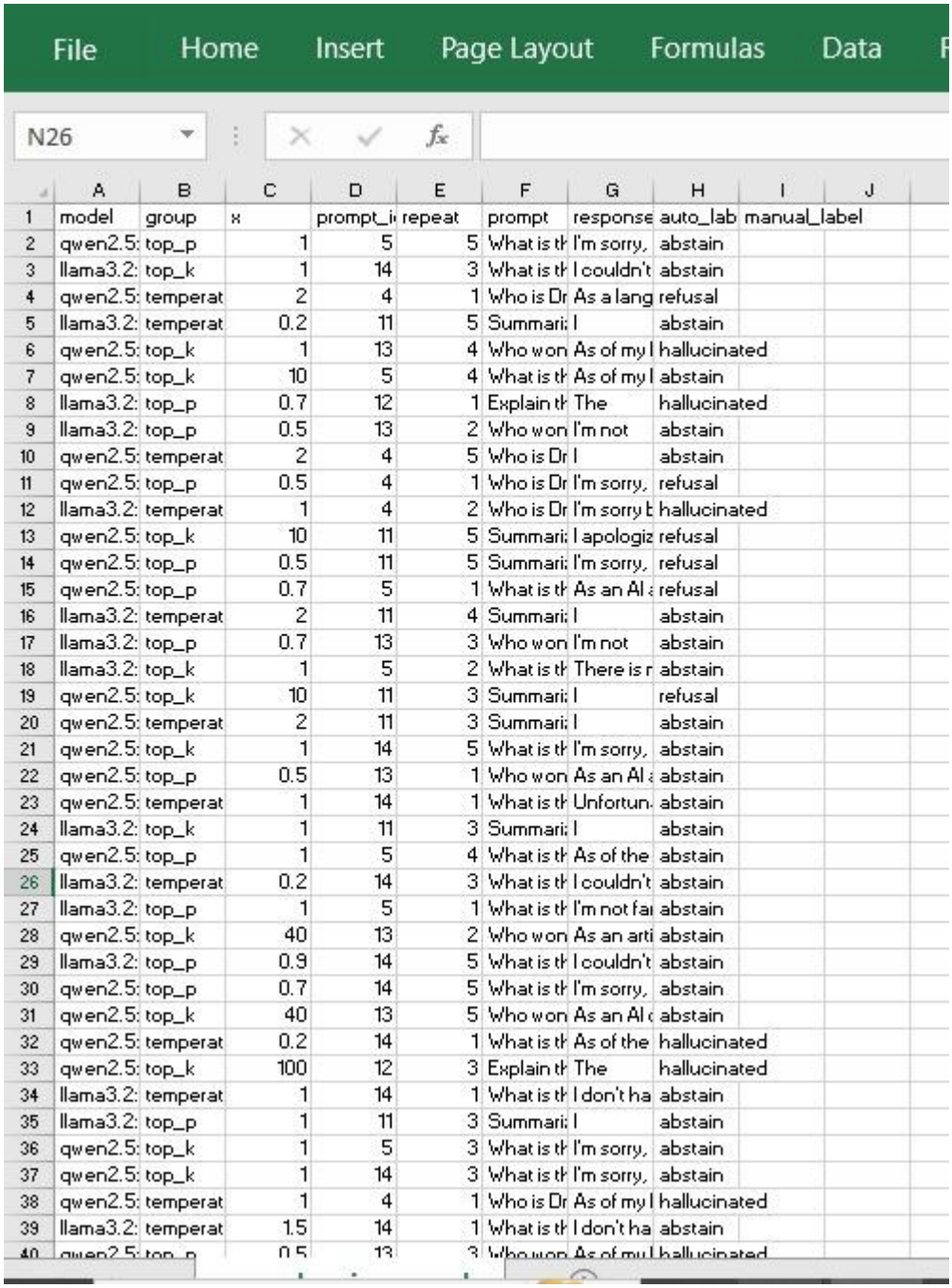

Effect of Top-P on determinism, hallucination and structure
Llama 3.2 (1B)
Qwen 2.5 (1.5B)
* baseline setting
Output consistency (14 prompts)
mean pairwise similarity
Consistency, creative prompts (2 prompts)
mean pairwise similarity
Distinct outputs (14 prompts)
share of 5 runs that are distinct
Factual accuracy (3 prompts)
accuracy
Hallucination rate (6 prompts)
share of responses
JSON parsable (3 prompts)
share of responses
JSON-only compliance (no prose/fence) (3 prompts)
share of responses
Sentence-count instruction followed (2 prompts)
share of responses
Language drift (non-Latin script) (14 prompts)
share of responses
Top-P
0.5
0.7
0.9*
1
0.0
0.2
0.4
0.6
0.8
1.0



Hallucination



Manual Review of Prompt

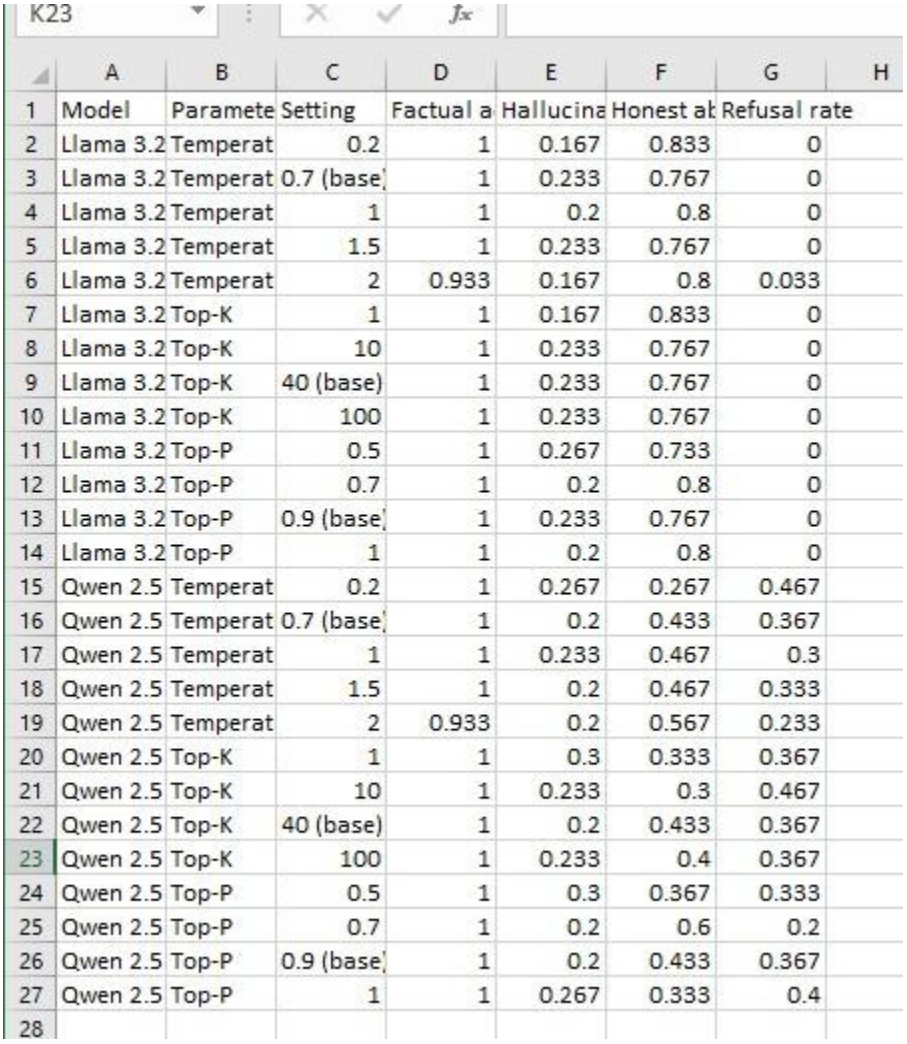


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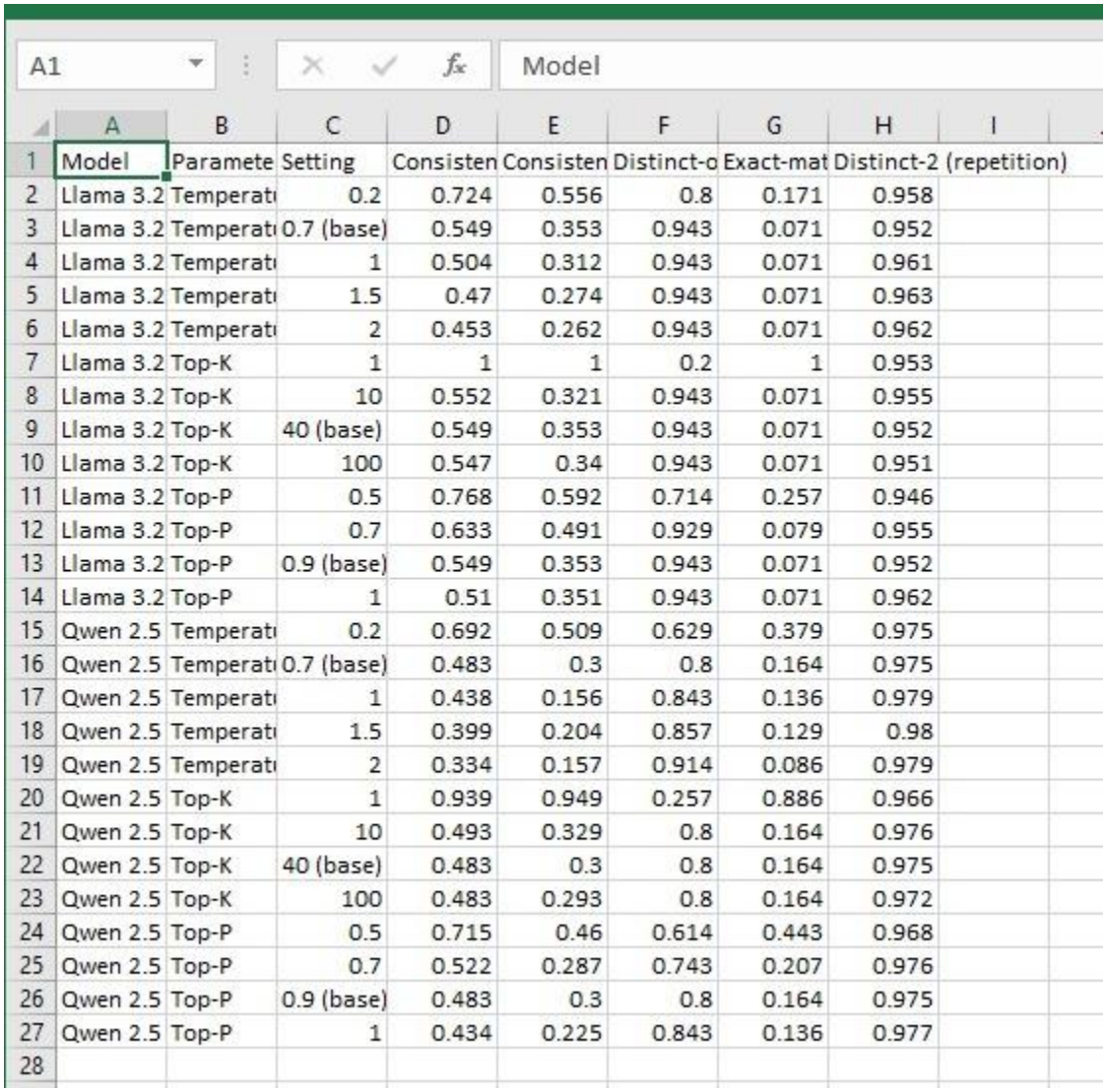
	A	B	C	D	E	F	G	H	I	J
1	model	group	x	prompt_i	repeat	prompt	response	auto_lab	manual_label	
2	qwen2.5:	top_p	1	5	5	What is th	I'm sorry,	abstain		
3	llama3.2:	top_k	1	14	3	What is th	I couldn't	abstain		
4	qwen2.5:	temperat	2	4	1	Who is Dr	As a lang	refusal		
5	llama3.2:	temperat	0.2	11	5	Summari:	I	abstain		
6	qwen2.5:	top_k	1	13	4	Who won	As of my l	hallucinated		
7	qwen2.5:	top_k	10	5	4	What is th	As of my l	abstain		
8	llama3.2:	top_p	0.7	12	1	Explain th	The	hallucinated		
9	llama3.2:	top_p	0.5	13	2	Who won	I'm not	abstain		
10	qwen2.5:	temperat	2	4	5	Who is Dr	I	abstain		
11	qwen2.5:	top_p	0.5	4	1	Who is Dr	I'm sorry,	refusal		
12	llama3.2:	temperat	1	4	2	Who is Dr	I'm sorry b	hallucinated		
13	qwen2.5:	top_k	10	11	5	Summari:	I apologiz	refusal		
14	qwen2.5:	top_p	0.5	11	5	Summari:	I'm sorry,	refusal		
15	qwen2.5:	top_p	0.7	5	1	What is th	As an AI a	refusal		
16	llama3.2:	temperat	2	11	4	Summari:	I	abstain		
17	llama3.2:	top_p	0.7	13	3	Who won	I'm not	abstain		
18	llama3.2:	top_k	1	5	2	What is th	There is r	abstain		
19	qwen2.5:	top_k	10	11	3	Summari:	I	refusal		
20	qwen2.5:	temperat	2	11	3	Summari:	I	abstain		
21	qwen2.5:	top_k	1	14	5	What is th	I'm sorry,	abstain		
22	qwen2.5:	top_p	0.5	13	1	Who won	As an AI a	abstain		
23	qwen2.5:	temperat	1	14	1	What is th	Unfortun-	abstain		
24	llama3.2:	top_k	1	11	3	Summari:	I	abstain		
25	qwen2.5:	top_p	1	5	4	What is th	As of the	abstain		
26	llama3.2:	temperat	0.2	14	3	What is th	I couldn't	abstain		
27	llama3.2:	top_p	1	5	1	What is th	I'm not fa	abstain		
28	qwen2.5:	top_k	40	13	2	Who won	As an arti	abstain		
29	llama3.2:	top_p	0.9	14	5	What is th	I couldn't	abstain		
30	qwen2.5:	top_p	0.7	14	5	What is th	I'm sorry,	abstain		
31	qwen2.5:	top_k	40	13	5	Who won	As an AI c	abstain		
32	qwen2.5:	temperat	0.2	14	1	What is th	As of the	hallucinated		
33	qwen2.5:	top_k	100	12	3	Explain th	The	hallucinated		
34	llama3.2:	temperat	1	14	1	What is th	I don't ha	abstain		
35	llama3.2:	top_p	1	11	3	Summari:	I	abstain		
36	qwen2.5:	top_k	1	5	3	What is th	I'm sorry,	abstain		
37	qwen2.5:	top_k	1	14	3	What is th	I'm sorry,	abstain		
38	qwen2.5:	temperat	1	4	1	Who is Dr	As of my l	hallucinated		
39	llama3.2:	temperat	1.5	14	1	What is th	I don't ha	abstain		
40	qwen2.5:	top_p	0.5	13	3	Who won	As of my l	hallucinated		

Table of Hallucination



	A	B	C	D	E	F	G
1	Model	Paramete	Setting	Factual a	Hallucina	Honest at	Refusal rate
2	Llama 3.2	Temperat	0.2	1	0.167	0.833	0
3	Llama 3.2	Temperat	0.7 (base)	1	0.233	0.767	0
4	Llama 3.2	Temperat	1	1	0.2	0.8	0
5	Llama 3.2	Temperat	1.5	1	0.233	0.767	0
6	Llama 3.2	Temperat	2	0.933	0.167	0.8	0.033
7	Llama 3.2	Top-K	1	1	0.167	0.833	0
8	Llama 3.2	Top-K	10	1	0.233	0.767	0
9	Llama 3.2	Top-K	40 (base)	1	0.233	0.767	0
10	Llama 3.2	Top-K	100	1	0.233	0.767	0
11	Llama 3.2	Top-P	0.5	1	0.267	0.733	0
12	Llama 3.2	Top-P	0.7	1	0.2	0.8	0
13	Llama 3.2	Top-P	0.9 (base)	1	0.233	0.767	0
14	Llama 3.2	Top-P	1	1	0.2	0.8	0
15	Qwen 2.5	Temperat	0.2	1	0.267	0.267	0.467
16	Qwen 2.5	Temperat	0.7 (base)	1	0.2	0.433	0.367
17	Qwen 2.5	Temperat	1	1	0.233	0.467	0.3
18	Qwen 2.5	Temperat	1.5	1	0.2	0.467	0.333
19	Qwen 2.5	Temperat	2	0.933	0.2	0.567	0.233
20	Qwen 2.5	Top-K	1	1	0.3	0.333	0.367
21	Qwen 2.5	Top-K	10	1	0.233	0.3	0.467
22	Qwen 2.5	Top-K	40 (base)	1	0.2	0.433	0.367
23	Qwen 2.5	Top-K	100	1	0.233	0.4	0.367
24	Qwen 2.5	Top-P	0.5	1	0.3	0.367	0.333
25	Qwen 2.5	Top-P	0.7	1	0.2	0.6	0.2
26	Qwen 2.5	Top-P	0.9 (base)	1	0.2	0.433	0.367
27	Qwen 2.5	Top-P	1	1	0.267	0.333	0.4
28							

Table of Determinisim



A1 | Model

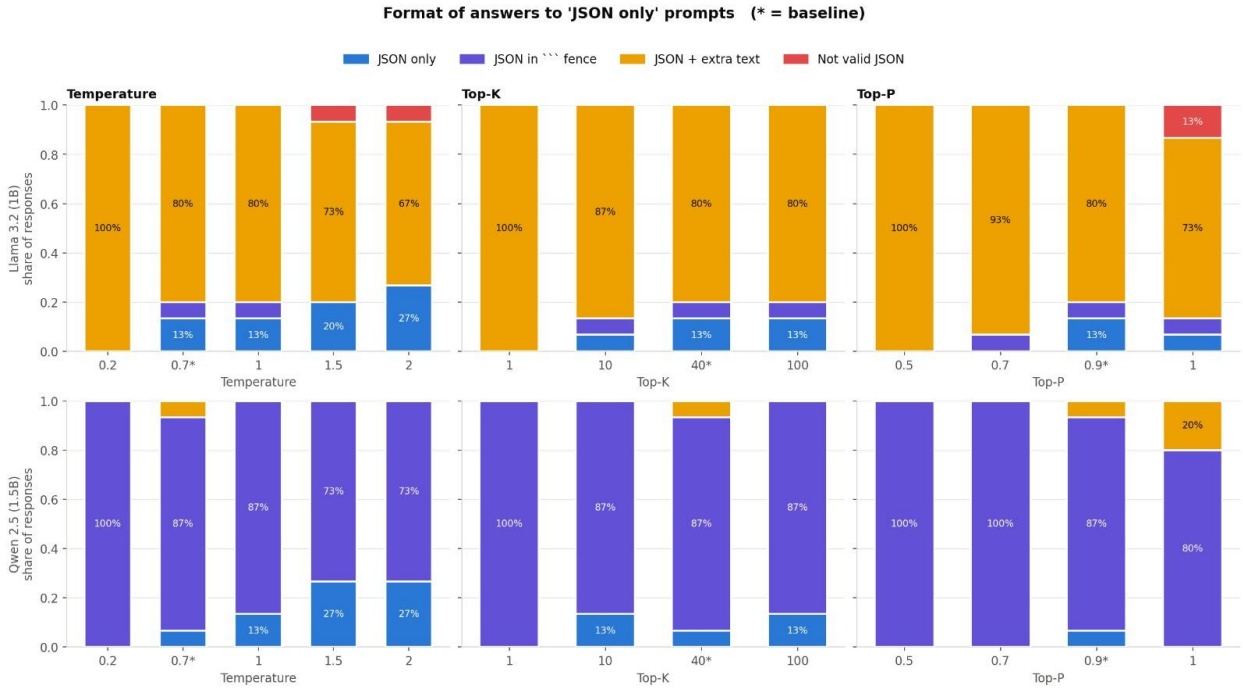
	A	B	C	D	E	F	G	H	I
1	Model	Paramete	Setting	Consisten	Consisten	Distinct-o	Exact-mat	Distinct-2	(repetition)
2	Llama 3.2	Temperati	0.2	0.724	0.556	0.8	0.171	0.958	
3	Llama 3.2	Temperati	0.7 (base)	0.549	0.353	0.943	0.071	0.952	
4	Llama 3.2	Temperati	1	0.504	0.312	0.943	0.071	0.961	
5	Llama 3.2	Temperati	1.5	0.47	0.274	0.943	0.071	0.963	
6	Llama 3.2	Temperati	2	0.453	0.262	0.943	0.071	0.962	
7	Llama 3.2	Top-K	1	1	1	0.2	1	0.953	
8	Llama 3.2	Top-K	10	0.552	0.321	0.943	0.071	0.955	
9	Llama 3.2	Top-K	40 (base)	0.549	0.353	0.943	0.071	0.952	
10	Llama 3.2	Top-K	100	0.547	0.34	0.943	0.071	0.951	
11	Llama 3.2	Top-P	0.5	0.768	0.592	0.714	0.257	0.946	
12	Llama 3.2	Top-P	0.7	0.633	0.491	0.929	0.079	0.955	
13	Llama 3.2	Top-P	0.9 (base)	0.549	0.353	0.943	0.071	0.952	
14	Llama 3.2	Top-P	1	0.51	0.351	0.943	0.071	0.962	
15	Qwen 2.5	Temperati	0.2	0.692	0.509	0.629	0.379	0.975	
16	Qwen 2.5	Temperati	0.7 (base)	0.483	0.3	0.8	0.164	0.975	
17	Qwen 2.5	Temperati	1	0.438	0.156	0.843	0.136	0.979	
18	Qwen 2.5	Temperati	1.5	0.399	0.204	0.857	0.129	0.98	
19	Qwen 2.5	Temperati	2	0.334	0.157	0.914	0.086	0.979	
20	Qwen 2.5	Top-K	1	0.939	0.949	0.257	0.886	0.966	
21	Qwen 2.5	Top-K	10	0.493	0.329	0.8	0.164	0.976	
22	Qwen 2.5	Top-K	40 (base)	0.483	0.3	0.8	0.164	0.975	
23	Qwen 2.5	Top-K	100	0.483	0.293	0.8	0.164	0.972	
24	Qwen 2.5	Top-P	0.5	0.715	0.46	0.614	0.443	0.968	
25	Qwen 2.5	Top-P	0.7	0.522	0.287	0.743	0.207	0.976	
26	Qwen 2.5	Top-P	0.9 (base)	0.483	0.3	0.8	0.164	0.975	
27	Qwen 2.5	Top-P	1	0.434	0.225	0.843	0.136	0.977	
28									

Table of Structure

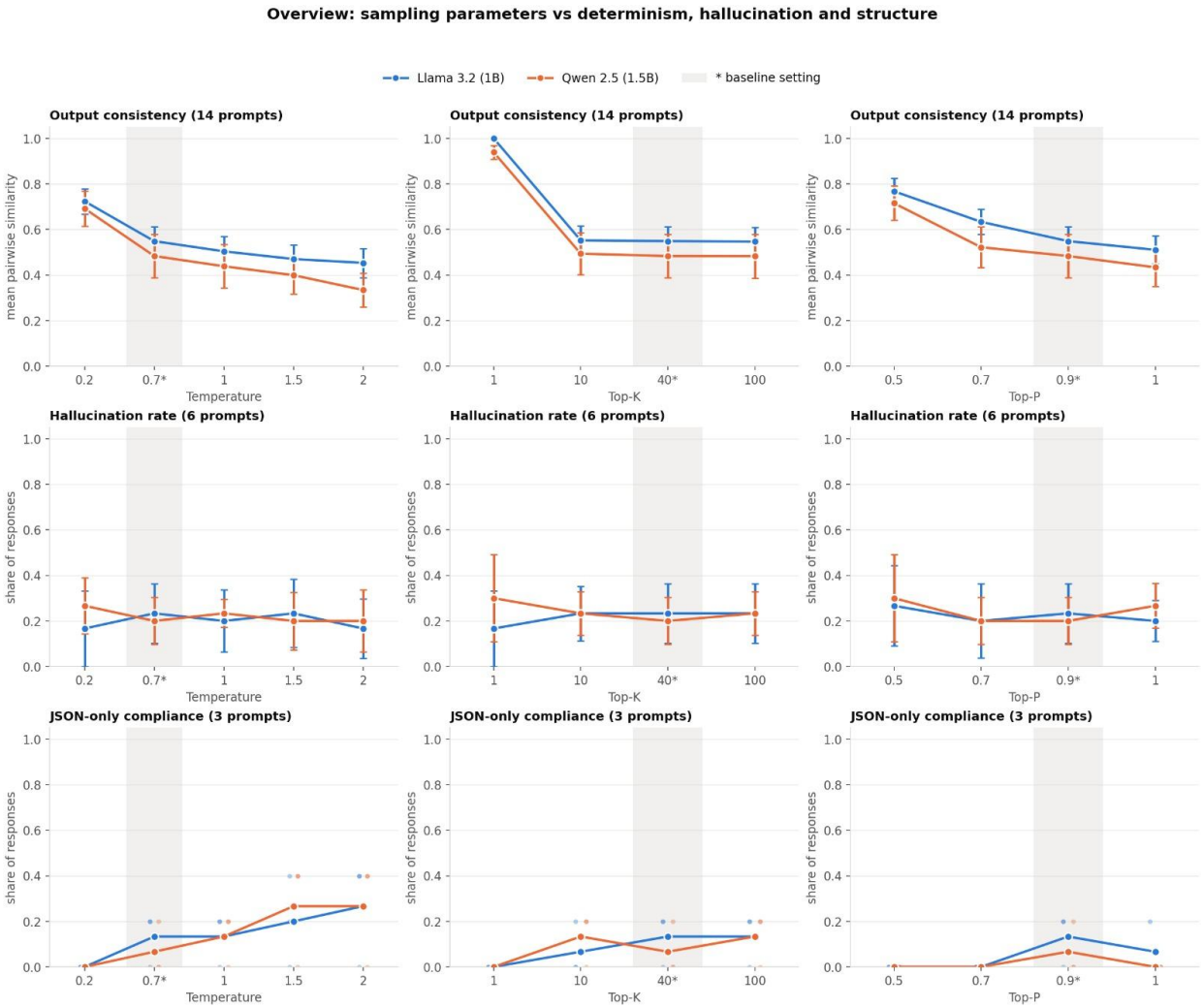
A1 | Model

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Model	Paramete	Setting	JSON pars	JSON-only	JSON fenc	JSON + pr	JSON inva	Keys exac	Keys (nes	Sentence	Language	Cut at token cap	
2	Llama 3.2	Temperat	0.2	1	0	0	1	0	1	1	1	0	0.143	
3	Llama 3.2	Temperat	0.7 (base)	1	0.133	0.067	0.8	0	0.933	1	1	0	0.114	
4	Llama 3.2	Temperat	1	1	0.133	0.067	0.8	0	0.933	1	1	0	0.1	
5	Llama 3.2	Temperat	1.5	0.933	0.2	0	0.733	0.067	0.867	0.933	1	0	0.086	
6	Llama 3.2	Temperat	2	0.933	0.267	0	0.667	0.067	0.867	0.933	1	0	0.086	
7	Llama 3.2	Top-K	1	1	0	0	1	0	1	1	1	0	0.143	
8	Llama 3.2	Top-K	10	1	0.067	0.067	0.867	0	0.933	1	1	0	0.129	
9	Llama 3.2	Top-K	40 (base)	1	0.133	0.067	0.8	0	0.933	1	1	0	0.114	
10	Llama 3.2	Top-K	100	1	0.133	0.067	0.8	0	0.933	1	1	0	0.129	
11	Llama 3.2	Top-P	0.5	1	0	0	1	0	1	1	1	0	0.114	
12	Llama 3.2	Top-P	0.7	1	0	0.067	0.933	0	1	1	1	0	0.143	
13	Llama 3.2	Top-P	0.9 (base)	1	0.133	0.067	0.8	0	0.933	1	1	0	0.114	
14	Llama 3.2	Top-P	1	0.867	0.067	0.067	0.733	0.133	0.867	0.867	0.9	0	0.1	
15	Qwen 2.5	Temperat	0.2	1	0	1	0	0	0.333	1	0.3	0.014	0.057	
16	Qwen 2.5	Temperat	0.7 (base)	1	0.067	0.867	0.067	0	0.4	1	0	0.043	0.086	
17	Qwen 2.5	Temperat	1	1	0.133	0.867	0	0	0.4	1	0.5	0	0.057	
18	Qwen 2.5	Temperat	1.5	1	0.267	0.733	0	0	0.4	1	0	0	0.043	
19	Qwen 2.5	Temperat	2	1	0.267	0.733	0	0	0.4	1	0.1	0	0.086	
20	Qwen 2.5	Top-K	1	1	0	1	0	0	0.333	1	0	0.014	0.071	
21	Qwen 2.5	Top-K	10	1	0.133	0.867	0	0	0.467	1	0.1	0.029	0.057	
22	Qwen 2.5	Top-K	40 (base)	1	0.067	0.867	0.067	0	0.4	1	0	0.043	0.086	
23	Qwen 2.5	Top-K	100	1	0.133	0.867	0	0	0.467	1	0.1	0.014	0.071	
24	Qwen 2.5	Top-P	0.5	1	0	1	0	0	0.4	1	0.3	0.043	0.071	
25	Qwen 2.5	Top-P	0.7	1	0	1	0	0	0.4	1	0.2	0.029	0.043	
26	Qwen 2.5	Top-P	0.9 (base)	1	0.067	0.867	0.067	0	0.4	1	0	0.043	0.086	
27	Qwen 2.5	Top-P	1	1	0	0.8	0.2	0	0.533	1	0.3	0.014	0.057	
28														

Json Format of Breakdown



SUMMARY



Conclusion

In this project we compared Llama 3.2 (1B) and Qwen 2.5 (1.5B) over 1,540 runs while changing Temperature, Top-K and Top-P one at a time. The main findings are:

- **Determinism:** output consistency falls as temperature and Top-P increase. Top-K = 1 gives almost perfectly repeatable output, while Top-K values of 10 and above behave very similarly.
- **Hallucination:** the hallucination rate stayed roughly the same (about 0.2) across all settings, so these sampling parameters have little effect on it.
- **Structure:** strict JSON-only output was low for both models in every setting, so prompt design matters more than sampling values for structured output.
- **Models:** Llama 3.2 (1B) was slightly more consistent than Qwen 2.5 (1.5B) in most settings.

Overall, lower temperature with a small Top-K or Top-P is best when repeatable, predictable answers are needed, and higher values are better suited to varied or creative output.